

Inverse rendering through a multiphysics equilibrium

A differentiable LWIR camera composed with a live Fortran/JAX/PyTorch cooling model
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Abstract. A thermal image is not temperature. It is band-integrated Planck radiance, modified by emissivity, reflected ambient, projection, optical blur, vignetting, and sensor calibration. We implement that measurement model as a differentiable JAX Tesseract and pull a pixel loss through it, through a coupled flow-heat fixed point, and back to an unknown volumetric heat source. The coupled system spans a PyTorch viscosity closure, a Fortran Darcy/Brinkman solver with a hand-derived adjoint, and JAX heat transport. The complete pixels-to-source derivative matches finite differences at $3.61\text{e-}07$ in-process and $1.14\text{e-}07$ through four served containers. Under independent truth/inversion grids, a modest camera-calibration mismatch remains within 2.14x of the calibrated pixel RMS yet moves the inferred hotspot an additional 1.82 coarse cells. The renderer therefore changes the physical diagnosis; it is not a visualization layer.

One noisy thermal image → the heat source that made it: gradient descent through a differentiable LWIR camera and a Fortran/JAX/PyTorch equilibrium

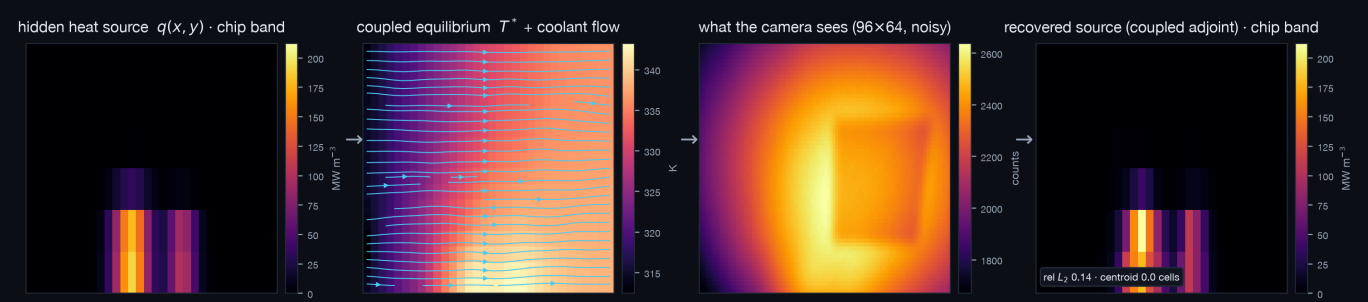


Figure 1. One noisy LWIR frame is inverted through the camera and the coupled equilibrium to recover the hidden heat-source map. Source panels share a physical scale.

Three contributions

Composition. Four typed Tesseracts expose apply and VJP endpoints across three differentiation systems. A matrix-free implicit adjoint composes them without a global tape.

Reproducibility. Fast CI verifies the API path. A separate target builds and serves all four images and stores image IDs, source commit, timing, and the finite-difference table.

Evidence. A committed-before-results renderer ablation uses a 64×32 truth grid and 32×16 inversion, reports negative controls, and makes calibration sensitivity measurable.

Honest limits. Same-grid accuracy, optimizer limits, synthetic data, and the no-PSF negative result are all reported. Claims are bounded to what the artifacts establish.

Evidence dashboard

API gradient	4-container gradient	same-grid source L2	calibration shift
3.61e-07	1.14e-07	0.1370	+1.82 cells

Track 05 thesis. The decisive result is not that a camera can be differentiated. It is that modest calibration error changes the recovered physical cause even when the image fit remains inside the predeclared plausibility envelope. Tesseract makes that camera VJP composable with an implicit, cross-language equilibrium.

One chain rule across four components

The source q drives a steady heat solve. Temperature changes viscosity; viscosity changes flow; flow changes temperature. At equilibrium $T^* = G(T^*; q)$. The camera maps T^* to digital counts $C = R(T^*)$. For a pixel loss J , the camera VJP supplies dJ/dT^* . The implicit adjoint solves $(I - dG/dT)^T \lambda = dJ/dT^*$ with GMRES, then one heat VJP returns dJ/dq . Each matrix-vector product is a chain of component VJPs; no framework traces the whole program.

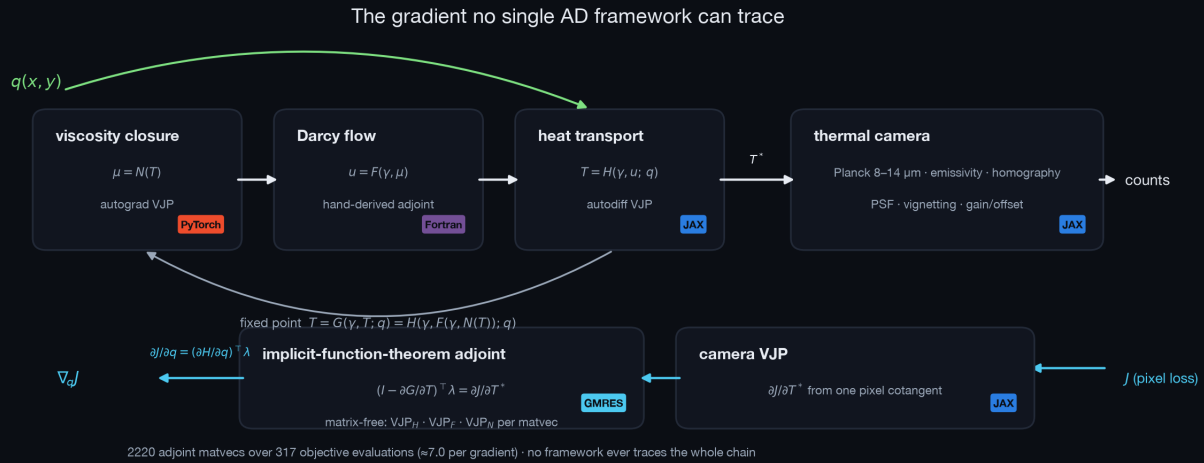


Figure 2. Forward fixed point and reverse implicit adjoint. The fast path uses identical APIs in-process; the interoperability gate serves the four images.

Thermal-camera Tesseract

The JAX renderer integrates Planck spectral radiance over 8-14 μm with 64-node Gauss-Legendre quadrature. A grey opaque surface emits $\epsilon L(T)$ and reflects $(1-\epsilon)L(T_{\text{ambient}})$. A homography and bilinear sampler project radiance to the sensor; a differentiable Gaussian PSF, $\cos^4$ vignetting, gain, and offset complete the image.

Heterogeneous physics

Viscosity is a learned PyTorch closure. Flow is a compiled Fortran Darcy/Brinkman solver with a hand-derived discrete adjoint. Heat transport is JAX with an implicit derivative. Only arrays and typed derivative requests cross component boundaries.

Verification hierarchy

gate	result
camera tests	18 pass
inverse utilities	7 pass
pixels-to-q FD, API	3.61e-07
pixels-to-q FD, 4 containers	1.14e-07
served runtime	80 s

Why Tesseract is load-bearing

JAX cannot trace the PyTorch tape or the Fortran binary; neither tape contains the equilibrium solve. Tesseract's common apply/VJP contract is the compositional abstraction. Removing it does not merely slow the system: it removes the cross-language derivative needed by the inverse problem.

The renderer changes the diagnosis

Experiment C was committed before its first result-producing run. Observations come from the complete camera over a 64x32 coupled solve; all recoveries invert at 32x16 with identical physics, data, TV prior, initialization, optimizer, and budget. Only the assumed camera changes.

A plausible thermogram fit can still give the wrong diagnosis — the renderer is load-bearing

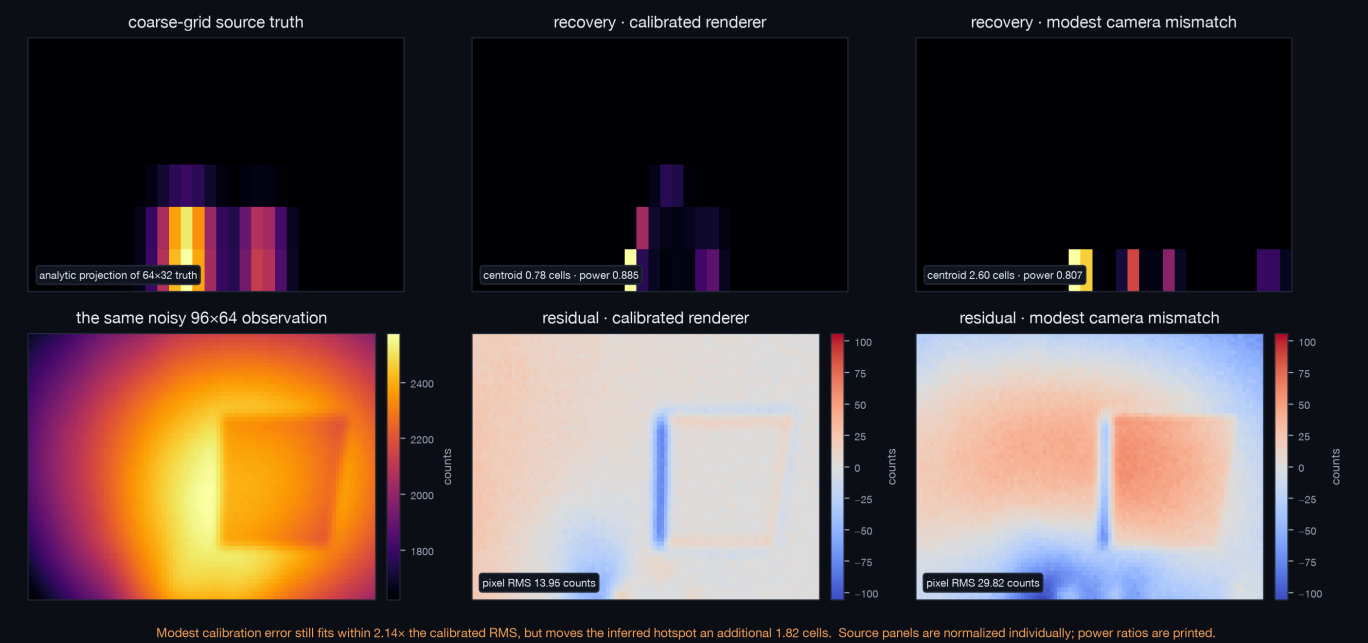


Figure 3. Calibrated and modest-mismatch recoveries see the same observation. Source panels are normalized individually; printed power ratios preserve amplitude information. The residual color scale is shared.

camera assumption	pixel RMS	source L2	centroid	power	status
calibrated	13.95	1.248	0.78	0.885	limit
blackbody	54.32	1.307	0.56	0.766	limit
no PSF	14.51	1.007	0.79	0.885	limit
no vignette	142.42	2.149	5.27	0.531	conv
modest mismatch	29.82	1.403	2.60	0.807	limit

Predeclared verdict: PASS. A simplified arm must fit within 3x the calibrated RMS and materially worsen source L2, centroid, or power. Only the modest calibration-mismatch arm passes both: 29.82 counts RMS and an additional 1.82-cell centroid error.

Negative evidence retained. Blackbody and no-vignetting models are visibly rejectable. Removing PSF blur fits nearly as well and improves coarse-grid source L2 from 1.248 to 1.007. The supported claim is calibration sensitivity, not that every optical stage is indispensable.

What the complete evidence establishes

Same-grid source recovery

Experiment B v2 recovers the two-hotspot map at relative L2 0.1370, centroid 0.02 cells, and power 1.000. A frozen-viscosity forward model ends at L2 0.2457. Both L-BFGS-B arms reach the 250-iteration limit; the comparison is a budget-matched model-mismatch endpoint, not a gradient-only intervention or convergence floor.

Independent-grid stress test

With 64x32 truth and 32x16 inversion, calibrated source L2 rises to 1.248 and pixel RMS to 13.95 counts. The old 0.1370 accuracy is therefore not robust to discretization mismatch. Centroid and power remain useful diagnostic summaries, but exact source shape is ill-conditioned from one blurred image.

Limitations

Measurements remain synthetic; one image underdetermines a 512-value source; calibration uncertainty is represented by one fixed mismatch rather than a posterior; material and camera physics are prototype-grade rather than instrument-grade; and four of five Experiment C optimizers stop at their budget. Real calibrated imagery, transient or multi-view data, held-out source families, and uncertainty maps are the next validation layer.

Reproduce

```
$ pip install -r requirements.txt
$ make judge
$ make verify
$ make verify-containers
$ make experiment-c
$ make figures renderer-figure landing
```

Container receipt

Mode: **served_containers**
Best FD error: **1.14e-07**
Runtime: 80 s
Source: 4031a389dbf
Four immutable image IDs are stored in the JSON artifact.

Artifact map

Protocol: writeup/RENDERER_PROTOCOL.md
Renderer result: figures/experiment_c_renderer.json
Container receipt: figures/container_e2e_gradient_check.json
Full narrative: writeup/WRITEUP.md
Code and data: github.com/il-miscusi/tesseract-inverse-thermography

Claim-to-artifact matrix

claim	direct evidence	boundary
Cross-framework derivative is correct	two end-to-end FD receipts	directional checks, small grids
Renderer calibration changes diagnosis	Experiment C, +1.82-cell shift	one fixed mismatch
Coupled model improves same-grid recovery	Experiment B v2, 0.137 vs 0.246 L2	both arms hit budget
Every optical stage is necessary	not supported: no-PSF is negative	claim rejected

Transfer applications

Electronics cooling. Localize a failing die or blocked channel from the visible cold-plate surface while respecting convection and radiometry.

Additive manufacturing. Pull pyrometer residuals through a differentiable sensor and melt-pool equilibrium to infer hidden heat input.

Active thermography. Infer delamination or subsurface defects from transient surface radiation with a swappable camera component.

References

[1] Pasteur Labs, *Tesseract: universal, autodiff-native software components*, 2026. [2] J. Bradbury et al., *JAX: composable transformations of Python+NumPy programs*, 2018. [3] A. Paszke et al., *PyTorch: An Imperative Style, High-Performance Deep Learning Library*, NeurIPS 2019. [4] P. Virtanen et al., *SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python*, Nature Methods 2020. [5] M. Planck, *On the Law of Distribution of Energy in the Normal Spectrum*, 1901.